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Development and simulation of two novel indoor odor source localization methods using a modified shark smell optimization algorithm

Qin Lin [a], Sihuan Wu [a], Sifan Wu [a], Hui Wang [a], Jinxiu Zhang [a,b,*]

[a] School of Aeronautics and Astronautics, Shenzhen Campus of Sun Yat-Sen University, Shenzhen, 518107, China
[b] Southern Marine Science and Engineering Guangdong Laboratory, Zhuhai, 519085, PR China

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ABSTRACT

Aiming to solve the problem of odor source localization (OSL) in the presence of interference sources, this paper presents two methods based on swarm intelligence algorithms. We initially introduced the shark smell optimization (SSO) algorithm and modified it for OSL tasks. Subsequently, mechanisms for collective information sharing and preventing falling into local minima were incorporated, leading to the development of the Improved Shark Smell Optimization (I-SSO) algorithm. We tested both algorithms in a computational fluid dynamics (CFD) simulated environment with a single interference source and compared them to the particle swarm optimization (PSO) algorithm and whale optimization algorithm (WOA). In scenarios with one and two interference sources. The results showed that the I-SSO algorithm outperformed the other three algorithms in both environment settings, demonstrating a higher success rate and superior search distance efficiency.

1. Introduction

The indoor environment is pivotal to human health, given that individuals spend upwards of 87 % of their time inside buildings [1 2]. Despite this, indoor spaces remain vulnerable to pollutants, including hazardous gas leaks, which pose significant health risks. Examples include the 1995 Tokyo subway sarin attack [3] and the COVID-19 pandemic. This imminent danger underscores the critical need for advanced research in gas leak source localization to promptly and accurately identify these threats. Traditionally, animals like dogs have been trained for OSL tasks to detect explosives or drugs [4], a method that is time-intensive, costly, and sometimes perilously unsafe for the animals in environments with toxic gases or fire. Consequently, developing autonomous robotic algorithms for OSL is not only innovative but vital for enhancing indoor safety and health.

Since the 1990 s, researchers have extensively studied robotic odor source localization, achieving significant breakthroughs [5]. However, accurately pinpointing leak sources in real-world practice remains a formidable challenge. Upon release, gases often encounter complex interactions with fluent airflow and obstacles, resulting in intricate concentration fields. Additionally, leaks, such as those from aging pipelines, seldom occur in isolation; minor cracks often accompany a primary leak, introducing interference gas sources. These disrupt the concentration field, creating local extremities that can easily mislead robots, ultimately causing the failure of localization tasks.

Existing approaches primarily divide the task of odor source localization into three sub-tasks [6]: plume finding, plume tracing, and source declaration. Current research primarily focuses on plume tracing, which is also the main focus of this paper. For the phase of plume tracing, existing methods can primarily be divided into five categories: gradient-based algorithms, bio-inspired algorithms, probabilistic algorithms, optimization algorithms and learning methods.

Early research primarily focused on gradient-based algorithms, which are relatively simple and easy to implement [7]. However, in complex turbulent environments where plumes exhibit meandering and intermittent characteristics, algorithms that climb concentration gradients do not always work effectively, especially in environments with interference sources, where they encounter significant difficulties. Bio-inspired algorithms are proposed based on the inspiration drawn from biological olfactory navigation behaviors, such as moth-inspired algorithms [8], lobster-inspired algorithms [9], and dung beetle-inspired algorithms [10]. The advantage of these algorithms is their low CPU resource consumption. However, they present challenges in evolving into multi-robot algorithms, as it is difficult to incorporate mechanisms

* Corresponding author at: School of Aeronautics and Astronautics, Shenzhen Campus of Sun Yat-Sen University, Shenzhen 518107, China. Southern Marine Science and Engineering Guangdong Laboratory, Zhuhai, 519085, PR China.

E-mail address: zhangjinxiu@sysu.edu.cn (J. Zhang).

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for inter-robot cooperation while maintaining their existing movement strategies [7].

Probabilistic algorithms treat the odor source's location as a probability distribution, and as the robot moves, it gains more and more environment knowledge, gradually moving toward the odor source. This category of algorithms includes infotaxis [11 12], hidden Markov methods [13], and standard Bayesian inference [14], among others. These algorithms offer the advantage of flexibility, allowing for the selection of different plume models based on varying scenarios [15]. However, considering that mobile robots have limited onboard computational resources, their high computational resource requirements limit their applicability in certain contexts, which are heavy burdens for hardware platforms. Moreover, these algorithms need a predefined gas diffusion model, and performance drops when this model deviates from reality.

Machine learning method was first introduced in [16] to address the issue of source declaration. Currently, with the development of reinforcement learning technology, an increasing number of studies are employing reinforcement learning-based methods to address the OSL problem. reinforcement learning reinforcement learning algorithms model the interaction between robots and their environment: the robot obtains rewards by performing actions, and its goal is to take actions that maximize cumulative rewards [17]. Chen et al. [18] employed a Deep Q-Network, using stacked historical measurement data as input and outputting the expected cumulative future rewards of the robot's actions. Zhao et al. [19] were the first to apply deep reinforcement learning to source localization, developing a particle clustering-deep Q-network. These algorithms require a great amount of training data, which is time-consuming and not easily feasible in practical OSL tasks.

Optimization algorithms conceptualize the concentration field as a dynamic function that varies with time and location, treating the OSL problem as an optimization issue of this dynamic function. Optimization-based methods typically employ multiple robots and utilize heuristic algorithms for search, such as PSO algorithm [20], teaching–learning-based optimization method [21], and whale optimization algorithms [22], among others. Compared to single-robot approaches, these multi-robot optimization algorithms have advantages in terms of efficiency and robustness. The characteristics of these algorithms make them suitable for OSL problems, attracting significant attention from researchers in recent years [23].

To address leak localization in environments with obstacles and minor odor interference, we introduced the SSO algorithm, which was proposed by Abedinia et al. in 2016 [24]. The SSO algorithm optimizes the objective function by mimicking the shark's behavior of tracking concentration gradients and rotational movement during foraging, demonstrating satisfactory results. To adapt the SSO for OSL tasks, we made adaptive improvements to it.

The original SSO algorithm, however, lacked a mechanism for information sharing among agents, missing out on the benefits of collective behavior. To overcome this limitation and boost efficiency, we integrated a collective information term into the SSO and introduced an adaptive mechanism for adjusting the range of rotational movements, culminating in the I-SSO algorithm. To the best of our knowledge, this is its first application in leak source localization tasks.

The main contributions of this paper are as follows:

(1) Treating the OSL task as a dynamic function optimization problem, we introduced the SSO algorithm and adaptively modified it to make it suitable for application in OSL tasks.
(2) A mechanism for sharing group information and a variance-triggered local extremum escape mechanism have been added to the SSO algorithm, resulting in the I-SSO algorithm.
(3) The two algorithms were evaluated in a scenario with one interference source; in a scenario with two interference sources, the two algorithms were compared with the PSO algorithm and the WOA.

The rest of this paper is organized as follows: Section 2 mathematically models the odor field and elucidates the concept of utilizing optimization algorithms to solve the problem of leak source localization, then introduces the SSO algorithm. Section 3 provides a detailed description of the robot and the mathematical model of the SSO algorithm. Section 4 explains how the SSO algorithm is applied to OSL tasks and is further improved to form the ISSO algorithm. Section 5 details the simulation environment and the results, followed by the conclusions.

2. Theoretical model

In indoor turbulent environments, the concentration field of odors is dominated by turbulent airflows. The concentration is a multivariate function of factors such as location and time and it can be considered as a multivariate function defined in the three-dimensional space, and its mathematical description is as follows:

$$c = F(\boldsymbol{X}, t) \tag{1}$$

where $\boldsymbol{X} = [x, y, z]$ is the position vector, and c is the odor concentration at position $\boldsymbol{X}$. t refers to the time at which the concentration is measured. The gas leak source localization problem can be conceptualized as a dynamic multivariate function issue, with its mathematical expression as follows:

$$\max_{\boldsymbol{X} \in \boldsymbol{G}} F(\boldsymbol{X}, t) \tag{2}$$

where $\boldsymbol{G}$ represents the distribution range of the concentration field. Consequently, it is amenable to resolution through swarm intelligence optimization algorithms.

People have been inspired by animal group behaviors to propose many swarm intelligence optimization algorithms. As an emerging optimization algorithm, the SSO algorithm has garnered attention from researchers due to its high exploration and exploitation capabilities and has been applied in various fields [25].

Sharks are animals with an exceptionally keen sense of smell, relying on their olfactory abilities to hunt. In the preying process, the shark proceeds to the prey based on the sense of smelling skills, and by increments in prey blood odor concentration, detects the prey. This behavior is referred to as forward movement. In addition to moving forward, sharks usually perform rotational movements along their path to find stronger odor particles and improve the direction of their progress.

The SSO algorithm emulates the predatory behavior of sharks, encompassing both forward movement and rotational movement. This dual approach enables the algorithm to effectively navigate various regions of the solution space, thereby exhibiting a strong exploration capability and search within these areas with high resolution.

3. Mathematical model

3.1. Mathematical model of the SSO algorithm

As illustrated in Fig. 1, the SSO algorithm is divided into forward movement and rotational movement. It solely utilizes concentration as a

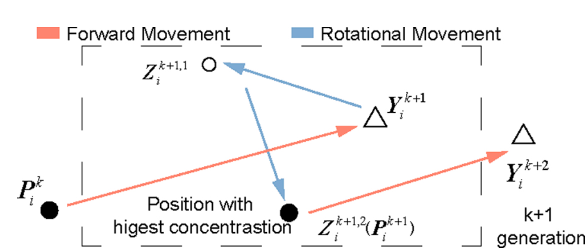

Fig. 1. Schematic of forward movement and rotational movement (M=2).

fitness function and does not rely on airflow information.

The time interval for updating the positions of shark swarms between each generation is fixed and denoted as Δt. The variable k represents the number of iterations, with each iteration encompassing both forward movement and rotational movement (noted that k does not represent the total number of iterations.

The shark follows odor and the direction of its movement is adjusted based on the odor intensity, and as the concentration of the odor increases, the velocity of the shark will increase. Also, the shark's current velocity depends on the previous velocity since it has inertia. The i^{th} shark's forward movement velocity is modeled as:

$$\boldsymbol{V}_i^k = c_1 R_1 \nabla(OF)|_{\boldsymbol{P}_i^k} + c_2 R_2 \boldsymbol{V}_i^{k-1} \tag{3}$$

where c_1 and c_2 belonging to [0, 1] are constants that determine the extent to which the concentration gradient and the previous velocity step influence the current velocity, with larger values indicating a greater degree of influence. R_1 and R_2 are random numbers with uniform distribution in the interval $[0, 1]$, which gives a more stochastic search nature to the algorithm. OF is the objective function and $\nabla(OF)|_{\boldsymbol{P}_i^k}$ indicates the gradient at position $\boldsymbol{P}_i^k$. The new position after forward movement, denoted as $\boldsymbol{Y}_i^{k+1}$, is given by the following equation:

$$\boldsymbol{Y}_i^{k+1} = \boldsymbol{P}_i^k + \boldsymbol{V}_i^k \Delta t \tag{4}$$

After completing the forward movement, sharks typically engage in a rotational movement to locate areas of higher concentration, which allows them to improve the direction of their process. This process is modeled as follows:

$$\begin{aligned} \boldsymbol{Z}_i^{k+1,m} &= \boldsymbol{Y}_i^{k+1} + \boldsymbol{r} \\ m &= 0, \ldots, M \end{aligned} \tag{5}$$

The parameter M is the number of points of the rotational movement, and by Eq. (5), the positions of rotational movement are generated in the vicinity of $\boldsymbol{Y}_i^{k+1}$.$\boldsymbol{r} = [r_x, r_y, r_z]$ is a random vector, where each component is independently and uniformly distributed in the interval [-D, D]. D represents the radius of activity for the rotational movement. If the shark detects a point with a stronger concentration during its rotational movement, it will move to a better position, determining the final location $\boldsymbol{P}_i^{k+1}$ for the $(k+1)^{\text{th}}$ iteration:

$$\boldsymbol{P}_i^{k+1} = \operatorname{argmax}\{F(\boldsymbol{Y}_i^{k+1}, t_k), F(\boldsymbol{Z}_i^{k+1,1}, t_k), \ldots, F(\boldsymbol{Z}_i^{k+1,M}, t_k)\} \tag{6}$$

Through the alternating progression of forward movement and rotational movement, the swarm of sharks continuously advances towards more optimal positions.

3.2. Robot model

Individual robots are assumed to be point masses without physical volume, disregarding their dynamics. This assumption accommodates a movement speed range of [0 m/s, 2 m/s] for the robots. In the simulation, it is assumed that each robot can determine its own position and the distances to various boundaries.

Consider a Cartesian coordinate system affixed to the terrestrial surface as reference frame A ($O_g - X_g Y_g Z_g$), and establish a right-handed coordinate system B ($O - XYZ$) on the robot platform. As shown in Fig. 2, the robot is assumed to be equipped with four gas concentration sensors, positioned on three vertical axes and at the center of the body. Sensor1 is at point O, Sensor2 is on the x-axis, Sensor3 is on the y-axis, and Sensor4 is on the z-axis. The three sensors on the axes are each at a distance Δh from Sensor1.

Since the robots are considered as point masses, their posture is not taken into account. During movement, the orientation of the robot's body coordinate system B relative to coordinate system A remains unchanged (the XYZ axes are parallel to the $X_g Y_g Z_g$ axes respectively). In three-dimensional space, the kinematic model of the robots is as follows:

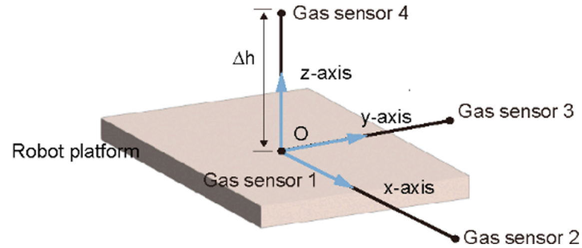

Fig. 2. schematic of robot model.

$$\boldsymbol{X}_{t+1} = \boldsymbol{X}_t + \boldsymbol{V}_t \Delta t \tag{7}$$

where $\boldsymbol{X}_t$ is the position vector of the robot at time t, $\boldsymbol{X}_{t+1}$ is the position vector of the robot at time t + 1, Δt is a fixed time interval, and Δt is the velocity vector given by the algorithm.

$$\boldsymbol{Y}_i^{k+1} = \boldsymbol{P}_i^k + \boldsymbol{V}_i^k \Delta t \tag{8}$$

4. Materials and methods

As with previous studies, we divided the source localization mission into three sub-tasks: plume finding, plume tracking, and source declaration. In this paper, the focus is on the plume tracking phase and the approach for the three stages is as follows:

(1) Plume finding phase: we use a random diverse search strategy. In the initial position, all robots randomly generate a forward direction vector and move along this direction at a constant speed. When any individual in the robot swarm detects a concentration greater than the threshold, all robots enter the plume tracking phase.
(2) Plume tracking phase: the key to the OSL algorithm lies in plume tracking. In this paper, we adopted two algorithms: SSO algorithm and I-SSO algorithm. The details of these two algorithms are described in Sections 2.1 and 2.2.
(3) Source declaration: When any individual in the robot swarm reaches the vicinity of the odor source, it is determined that the odor source has been successfully located.

The framework of the two proposed methods for source localization is illustrated in Fig. 3.

4.1. Source localization based on the SSO algorithm

We treat each robot as a shark and a group of robots as a shark group and do not consider its dynamics since the focus is on the navigation algorithm. In real-world situations, the concentration field is often intermittent and multi-peaked, making it challenging to obtain derivatives. Therefore, in this context, we use the concentration differences across three vertical axes on the robot itself (see Fig. 2) to substitute for the concentration gradient in Eq. (3), which has been modified as follows:

$$\boldsymbol{V}_i^k = c_1 R_1 (\Delta \boldsymbol{c} / \Delta h) + c_2 R_2 \boldsymbol{V}_i^{k-1} \tag{10}$$

where $\Delta \boldsymbol{c} = [\Delta c_x, \Delta c_y, \Delta c_z]$ and the variable Δc_x is the difference between the readings of sensor 2 and sensor 1, Δc_y is the difference between sensor 3 and sensor 1, and Δc_z is the difference between sensor 4 and sensor 1. Eq. (10) provides the specific adaptation of the SSO algorithm for OSL tasks, and the subsequent process of the SSO algorithm is the same as described in Section 3.1.

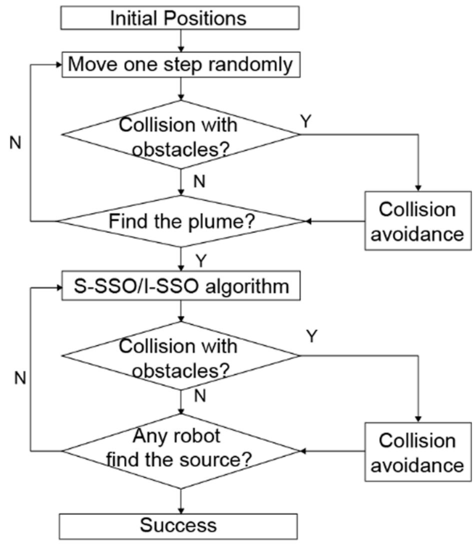

Fig. 3. Framework of source localization method based on the two proposed algorithms.

4.2. Source localization based on the I-SSO algorithm

In the SSO algorithm, each agent acts independently and does not interact with others. To more efficiently utilize collective information, we have incorporated the location information of the individual with the highest odor concentration into Eq. (4), adding a velocity component directed towards the optimal individual. In the standard SSO algorithm, each agent serves as an individual and does not contact each other. Eq. (10) is modified as follows:

$$\boldsymbol{V}_i^k = c_1R_1(\Delta\boldsymbol{c}/\Delta h) + c_2R_2\boldsymbol{V}_i^{k-1} + \gamma(\boldsymbol{P}_i^k - \boldsymbol{P}_{best}^k)/\left|(\boldsymbol{P}_i^k - \boldsymbol{P}_{best}^k)\right| \tag{14}$$

Where γ is the constant which determines the effect of collective information and $\boldsymbol{P}_{best}^k$ is the position vector of the individual with the highest concentration value among the k^{th} iteration.

Due to the turbulent nature of the odor plume, gases tend to accumulate in certain areas, forming local high concentration values. Additionally, the presence of interference sources can also create local maxima. To avoid being trapped in local maxima, we escape from this region by expanding the range of rotational movement.

Firstly, the robotic swarm needs to determine whether it has fallen into a local extremum. When the swarm is caught in a local extremum, the distances between individual robots tend to be very close. Therefore, in this paper, we assess the possibility of being in a local extremum by calculating the variance of their positions. The determination method is as follows:

$$S_x^i = \sum_i^n \left(x_i^k - \sum_i^n (x_i^k)/n\right)^2 \tag{15}$$

$$S_y^i = \sum_i^n \left(y_i^k - \sum_i^n (y_i^k)/n\right)^2 \tag{16}$$

$$S_z^i = \sum_i^n \left(z_i^k - \sum_i^n (z_i^k)/n\right)^2 \tag{17}$$

where S_x^i, S_y^i and S_z^i represent the variances of the three components of the robot's position coordinates and n is the number of robots. In each iteration, calculate the mean square deviation of the robotic swarm's positions. When it is less than a threshold value, modify the magnitude of the vector $\boldsymbol{r}$ in Eq. (6):

$$\begin{array}{l} r_x, r_y, r_z \in [-D_{escape}, D_{excape}],\ \text{if } S_x^i < S_{x,th}, S_y^i < S_{y,th}, S_z^i < S_{z,th} \\ r_x, r_y, r_z \in [-D, D],\ \text{if } S_x^i \geq S_{x,th}, S_y^i \geq S_{y,th}, S_z^i \geq S_{z,th} \end{array} \tag{18}$$

D_{escape} represents the range of the rotational movement when the group is in a state of aggregation. By altering the range of rotational movement, the robotic swarm can escape the current local concentration extremum and explore new information, thereby increasing the success rate of localization.

4.3. Obstacle avoidance algorithm

We have improved upon the obstacle avoidance algorithm previously proposed in [26], enhancing the 'reflex distance' of the robot when encountering obstacles. This modification helps to prevent the robot from being trapped in local extremum areas near walls or the edges of obstacles. The schematic of the obstacle avoidance algorithm is shown in Fig. 4. Assuming that the robot is equipped with ultrasonic sensors capable of detecting the distance to obstacles in its forward path when it is detected that the next movement step would result in a collision with an obstacle, the speed of forward movement $\boldsymbol{V}_i^k$ is modified as follows:

$$\boldsymbol{V}_i^{k'} = \boldsymbol{V}_i^k - (\boldsymbol{V}_i^k \cdot \boldsymbol{N} - d)\cdot\boldsymbol{N} \tag{19}$$

where d is a constant greater than 0 and $\boldsymbol{N}$ is the direction vector of the obstacle surface (pointing outward from the obstacle). The larger the value of d, the greater the distance between the robot and the obstacle after executing the obstacle avoidance algorithm. The new position after forward movement is modified as:

$$\boldsymbol{Y}_i^{k+1} = \boldsymbol{P}_i^k + \boldsymbol{V}_i^{k'}\Delta t \tag{20}$$

5. Simulations and results

In this section, we used commercial computational fluid dynamics software FLUENT to simulate the concentration distribution and then conducted simulation experiments in Matlab R2022a. The performance of the SSO and I-SSO algorithms was evaluated in the environment with one single interference source. Furthermore, a comparative analysis was conducted with the traditional PSO algorithm and WOA in environments with one and two interference sources, under conditions of random initial positions.

5.1. Environment settings

There are two settings in this paper: Setting I has one interference source (interference source1), and Setting II has two interference sources (interference source1 and interference source2), as shown in Fig. 5 (a) and (b). Both Setting I and Setting II utilize the same room for simulation and the room has a length of 20 m (x-axis), a width of 16 m (y-axis), and a height of 6 m (z-axis).

The RNG k-ε model was used to simulate the turbulent flow, which

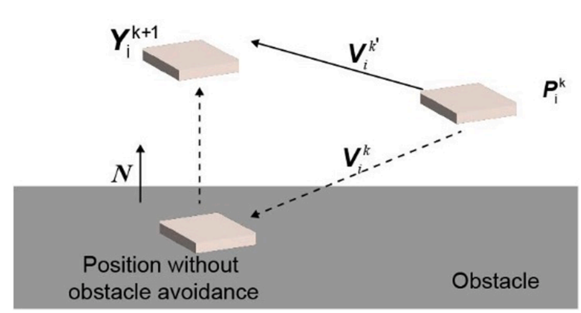

Fig. 4. Schematic of obstacle avoidance.

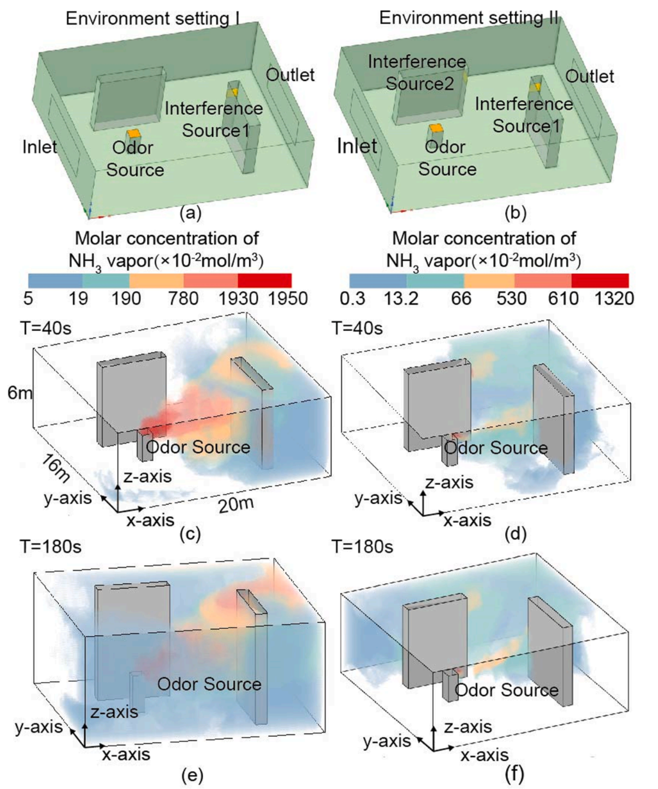

Fig. 5. The model of the testing environment and the concentration distribution map.(a) is the geometric model of setting I. (b) is the geometric model of setting II. (c) and (e) are the distribution of NH_3 vapor of setting I of time intervals 40 s and 180 s. (d) and (f) are the distribution of NH_3 vapor of setting II of time intervals 40 s and 180 s.

has good accuracy for many flows and was used to successfully simulate the indoor pollution distributions [27]. The gas leaking is ammonia vapor (NH3) and the release rate of the odor source is 0.5 g/s and 0.1 g/s of interference source 1 and 2. The odor source emits at a constant rate of 1 g/s, and the wind is stable with a velocity of 2 m/s. Fig. 5(c)–(e) illustrates the distribution of NH3 vapor in Setting I and Setting II of time intervals 40 s and 180 s.

5.2. Evaluation metrics

One successful localization to the odor source is determined when any robot moves into a 1 m^3 area near the odor source within 300 steps. We identified two metrics to evaluate the performance of the algorithm: (a) success rate, and (b) search distance. The success rate is defined as the ratio of the number of successful odor source localizations to the total number of experiments conducted under the same initial positions and environmental settings. The search distance refers to the average movement distance of the robot swarm in a successful experiment.

5.3. Performance of the SSO algorithm with one single interference source

This section investigates the performance of the SSO algorithm under various initial positions and parameter M in setting I. We utilized five robots for the search. The range of rotational movement, D, is 2 m. In practical tasks, robots operate without prior knowledge of the odor source's location, and their relative positions to it are unknown. Therefore, we selected three representative initial position distribution patterns. The coordinates of the initial positions are shown in Table 1. and Fig. 6.

Initial Position A: Evenly distributed throughout the space.

Initial Position B: Concentrated in areas of low concentration.

Table 1
The coordinates of the initial positions.

Initial Position	A	B	C
Coordinate	[2 6 2] [8 14 2] [12 2 2] [18 4 2] [18 12 2]	[2 6 2] [2 12 2] [2 14 2] [8 14 2] [4 4 2]	[16,12,4] [16,12,5] [16,11,4] [16,11,5] [17,12,4]

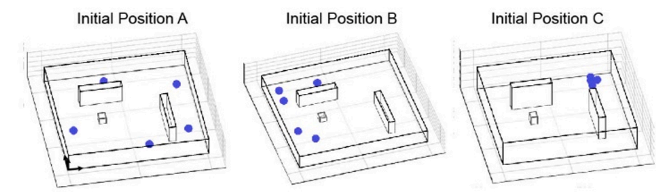

Fig. 6. Illustration of initial positions A, B and C.

Initial Position C: Concentrated near the leak source and the interference source.

Fig. 7 presents the statistical results of the SSO algorithm. The blue dots in the figure represent the initial positions of the robots. When the parameter M equals 0, it implies that the robots only perform forward movement, which is equivalent to five robots conducting olfactory searches based on concentration gradients and previous velocity simultaneously. As observed, the success rates in the three initial positions are only 47 %, 50 %, and 47 %, respectively. This is consistent with the conclusions from previous literature that tracking the plume solely based on concentration gradients is an extremely challenging task [7]. Movement along concentration gradients often leads to entrapment in local areas of high concentration, leading to the failure of the task. When the rotational movement is introduced (M=1), there is a significant increase in the success rate. Compared to searches based solely on concentration gradients, the success rates at initial position A, B and C improved by 43 %, 22 % and 41 %. We believe this improvement is attributable to the enhanced local search capabilities of the robots in the SSO algorithm, which are facilitated by rotational movements, thereby avoiding entrapment in the local extremum. The success rate of the SSO algorithm, after incorporating rotational movement, was at least 72 %, demonstrating its effectiveness in scenarios with a single interference source.

A clear trend observed at initial position A and C is that the success rate gradually increases with the increment of M, eventually plateauing. At initial position B, the trend first increases and then decreases. This

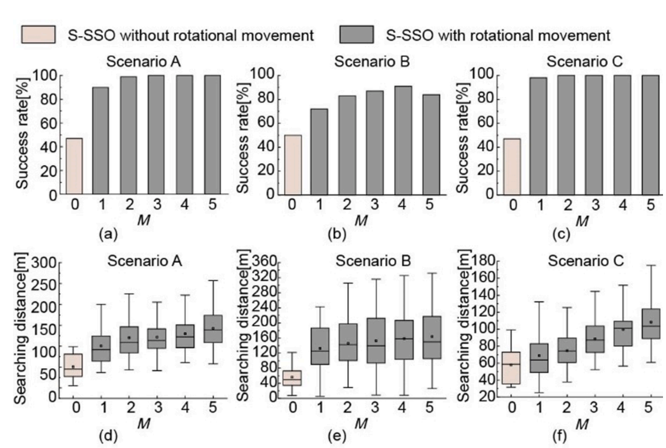

Fig. 7. Statistical results of SSO algorithm.(a) (b) and (c) are the success rate of the SSO algorithm in scenario A, B and C. (d) (e) and (f) are the search distance of the SSO algorithm in scenario A, B and C.

could be attributed to the experimental setup where the maximum number of iterations was set to 300 (including both rotational and forward movements). An excessive number of rotational movements may lead to an overemphasis on local searches at the expense of global exploration, resulting in a reduced success rate. Meanwhile, the search distance consistently increases as M increases. Clearly, as the number of local searches increases, the search distance also gradually increases.

Considering the success rate and search distance across three initial positions in relation to the variation in the number of local searches, it is evident that an increased frequency of rotational movements does not enhance the success rate. Instead, it leads to an increased search distance, thereby reducing the efficiency of the search.

5.4. Performance of I-SSO algorithm with one single interference source

Using the same environmental settings and initial positions, we evaluated the I-SSO algorithm's performance, with each parameter undergoing 100 experiments. The range of rotational movement, D, is the same as section 3.3 and D_{escape} equals 1. Fig. 8 presents the statistical results of the I-SSO algorithm. To more clearly demonstrate the enhancement of the algorithm's effectiveness due to the incorporation of collective information, we have included the statistical results of the SSO algorithm when M equals 0 (gradient-based search only) in the figure.

Compared to relying solely on concentration gradients, the inclusion of collective information alone, without the addition of rotational movement (I-SSO, M=0), also significantly enhances the algorithm's performance. At initial position A, the success rate increased by 42 %, and the search distance decreased by 40 %; in scenario B, the success rate improved by 3 %, and the search distance reduced by 44 %; in scenario C, the success rate rose by 42 %, and the search distance declined by 57 %. It is evident that the incorporation of collective information not only substantially boosts the success rate but also significantly reduces the search distance. This enhancement, similar to adding rotational movements, provides more than just gradient navigation, preventing robots from being trapped in local extremums and boosting success rates.

At initial positions A and C, the success rate of the I-SSO algorithm remains around 100 % with the increase of M. In contrast, at initial position B, the success rate initially shows a significant rise at$M$ = 3 reaching a peak of 81 %, and then slightly declines. However, even at its lowest success rate of 71 % (I-SSO, M=1), the outcome is still considered relatively satisfactory. This indicates that the I-SSO algorithm is effective in in scenarios with a single interference source. Similar to the SSO algorithm, the decrease in success rate in scenario B with the increase of

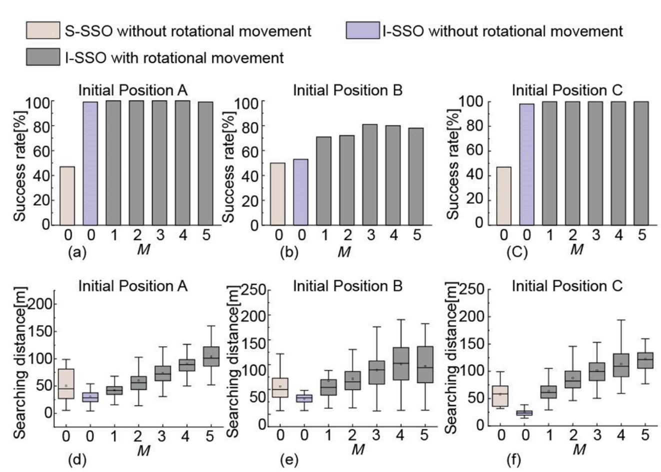

Fig. 8. Statistical results of the SSO algorithm.(a), (b) and (c) are the success rates of the I-SSO algorithm and SSO algorithm with parameter M equal to 0 in scenario A, B and C. (d) (e) and (f) are the search distances of the I-SSO algorithm in scenario A, B and C.

M could be attributed to the limitation of the total number of steps. Regarding the search distance, a clear trend is observed: as M increases, the search distance gradually increases.

In the I-SSO algorithm, when M equals 1, it signifies the simultaneous introduction of rotational movement and the mechanism for collective information sharing. The concurrent implementation of these two mechanisms enhances the robustness of the algorithm. Although in scenarios A and C, the introduction of one rotational movement in the I-SSO algorithm (I-SSO, M=1) does not significantly enhance the success rate compared to the I-SSO algorithm without Rotational Movement (I-SSO, M = 0), in scenario B, the concurrent introduction of both mechanisms achieves a success rate of 71.00 %, marking an improvement of 18 %. Therefore, the simultaneous introduction of both mechanisms, taking into account both the success rate and the search distance, proves to be a favorable choice. The concurrent implementation of these two mechanisms enhances the algorithm's robustness to different initial positions.

5.5. Comparison with other swarm intelligent algorithms

This section compares the performance of the two proposed algorithms with the PSO algorithm and WOA in environmental settings I and II. Both the PSO algorithm and WOA are bio-inspired swarm intelligence algorithms with applications across many fields. The WOA, introduced by Seyedali et al. in 2016 [28], has also been applied in the field of source localization [29]. As a newly emerging algorithm, the WOA has demonstrated excellent exploration abilities and has outperformed the PSO algorithm in escaping from local optima [28]. This aligns with the challenges that the presence of interference sources, which often create local optima, this paper aims to address.

Three sets of initial positions in each environment setting were generated (Initial positions D, E and F for setting I; Initial positions G, H and I for setting II). For each of these initial positions, we conducted 100 trials for each algorithm. The initial positions are shown in Table 2. and Table 3., and the parameters for the two proposed algorithms are detailed in Table 4.. For the PSO algorithm, we employed standard parameters: ω=0.8, l_1=0.5, andl_2 = 0.5.

Fig. 9 shows the success rate and search distance results in environment setting I. Both the SSO and I-SSO algorithms achieve high success rates across three initial positions, with a minimum success rate of 70 % and a maximum of 100 %. Among them, the I-SSO algorithm exhibited the best performance, achieving a 100 % success rate at initial positions D and F, and also attaining a success rate of 74 % at initial position E. Additionally, the I-SSO algorithm demonstrated the highest success rate at all three initial positions compared to the other two algorithms. Conversely, the success rates of the PSO algorithm and WOA were relatively low in all three scenarios, with the highest being 35 %. In most scenarios where failure occurred, the robot swarm was predominantly attracted to the vicinity of the interference source.

In terms of search distance, the I-SSO algorithm, benefiting from its adaptive rotational movement radius, demonstrated relatively shorter distances. At initial position D, the I-SSO algorithm reduced the search distance by 61 % compared to the SSO algorithm, and by 40 % compared to the PSO algorithm, which is 62 % less than the reduction achieved by the WOA. At initial position E, the reductions were 46 % compared to the SSO algorithm and 6 % compared to WOA. At initial position F, the

Table 2
Initial positions for setting I.

Initial positions	D	E	F
Coordinate	[3 7 5] [16 15 14] [13 5] [13 12 4] [8 10 1]	[8 1 5] [1 9 5] [5 8 4] [6 8 5] [7 4 5]	[5 5 1] [12 6 5] [3 5 4] [12 10 4] [16 13 1]

Table 3
Initial positions for setting II.

Initial positions	G	H	I
Coordinate	[5 14 13] [17 14 5] [14 8 2] [11 6 1] [14 4 5]	[12 2 3] [14 1 3] [18 5 3] [1 2 4] [17 12 2]	[4 9 4] [18 10 2] [18 2 1] [17 4 5] [1 1 3 5]

Table 4
Parameters of the SSO and I-SSO algorithm.

Algorithm	M	α	β	γ	D	D_{escape}
SSO	1	0.5	0.8	nan	2	nan
I-SSO	1	0.5	0.8	0.5	2	1

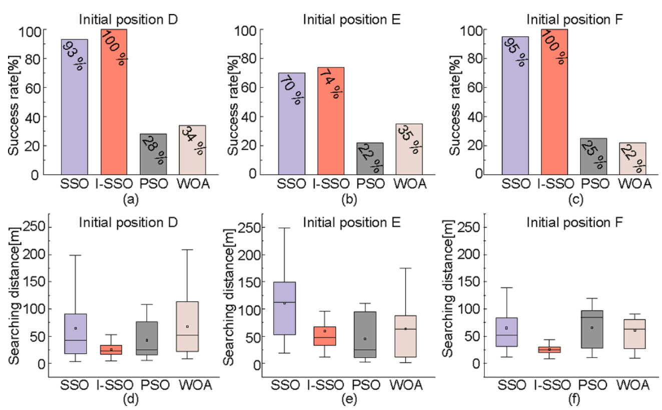

Fig. 9. Statistical results for SSO, I-SSO, PSO and WOA algorithms in three randomly generated initial positions in environment setting I.

reductions were 61 % compared to both the SSO and PSO algorithms, and 58 % less than the reduction achieved by WOA. Regarding both success rate and search distance, the I-SSO algorithm consistently outperforms the other three algorithms in environment setting I.

When the number of interference sources increased to two, there was a noticeable change in the algorithm's performance. Fig. 10 presents the statistical results in environment setting II. The success rates of the SSO, PSO algorithms, and WOA significantly decreased at all three initial positions, with the SSO algorithm achieving only 35 %, 20 %, and 17 % at initial positions G, H, and I, respectively, and the PSO algorithm at 10 %, 2 %, and 1 %. The WOA algorithm recorded success rates of only 24 %, 20 %, and 19 % respectively. In scenarios with two interference sources, the success rates of these three algorithms are not satisfactory.

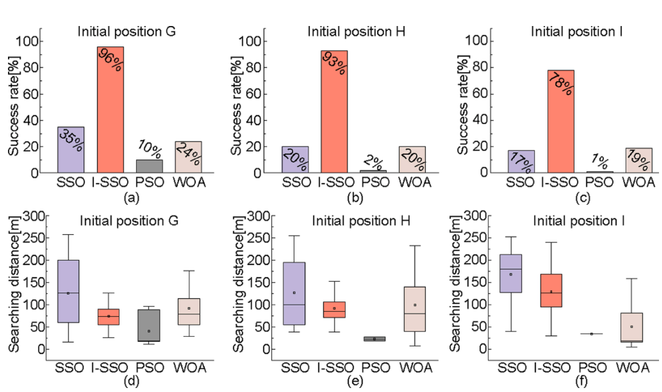

Fig. 10. Statistical results for SSO, I-SSO, PSO and WOA algorithms in three randomly generated scenarios.

In contrast, the I-SSO algorithm maintained a relatively high success rate, with 96 %, 93 %, and 78 % at initial positions G, H and I. Compared to the SSO algorithm, the I-SSO algorithm not only surpasses in success rate but also excels in search distance. At initial positions G, H and I, the search distance decreased by 42 %, 27 %, and 23 %, respectively.

Fig. 11 illustrates the process of the I-SSO algorithm successfully locating the odor source in environment setting II, with the robots' initial position at G. The points in different colors represent different robots. At t = 1 s, the robots (R^1 $-R^5$) are at their starting positions, beginning the search. By t = 31 s, robots R^1 and R^2 gradually approach interferences 1 and 2, while the remaining three robots are attracted by the plume generated by the odor source. At t = 41 s, R1 and R2 become trapped in the local extremum area produced by the interference. By t = 51 s, the highest concentration is observed between R^3, R^4, and R^5. Due to the collective information sharing mechanism of the I-SSO algorithm, R^1 and R^2 gradually move toward the other robots. At t = 61 s, all five robots move along the plume generated by the odor source. At t = 86 s, the odor source is successfully located.

6. Conclusions

In this paper, we have designed two novel swarm intelligence algorithms for locating gas leak sources in indoor environments with minor leak interference. Based on the results obtained in the CFD simulation environment, the main conclusions of the paper are as follows:

1) The introduction of rotational movement and collective information, as opposed to solely relying on concentration gradients for navigation, can enhance the success rate and reduce the search distance of the algorithm. The simultaneous incorporation of both elements demonstrates stronger robustness across various initial positions and different numbers of interference sources.
2) The I-SSO algorithm outperformed the other two algorithms in both environment settings, with a higher success rate: in the single interference source scenario (I-SSO: 95 %, SSO: 85 %, PSO: 18 %, WOA: 30 %) and in the double interference source scenario (I-SSO: 89 %, SSO: 24 %, PSO: 4 %, WOA: 21 %).

Future work will focus on testing the algorithm's performance in more realistic simulation environments. We will attempt to consider real sensor models, the robot's kinematic model, and sensor data processing,

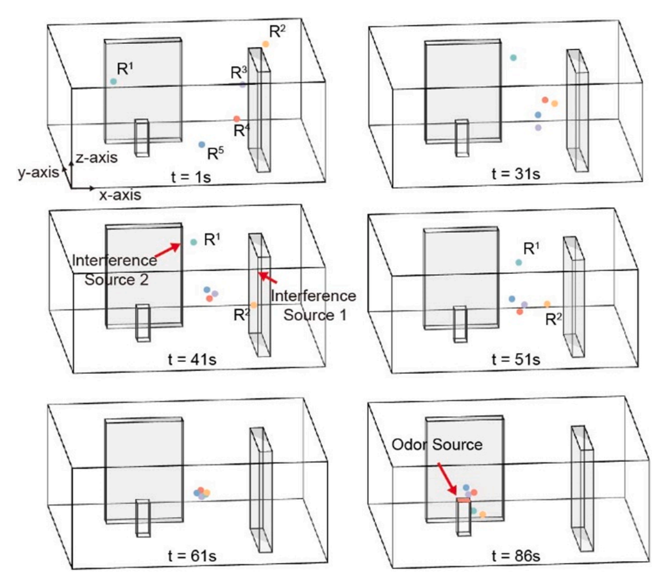

Fig. 11. A successful source localization process of the I-SSO algorithm in environment setting II.

among other factors.

CRediT authorship contribution statement

Qin Lin: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sihuan Wu:** Software, Investigation. **Sifan Wu:** Visualization, Software. **Hui Wang:** Writing – review & editing, Methodology. **Jinxiu Zhang:** Conceptualization, Resources, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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